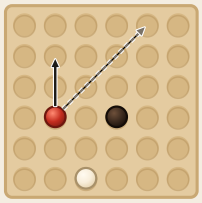
DORON

Tactics and strategy, illustrated on the 6 × 6 game

Every position in this guide comes from a real game between computer players and has been **solved exactly to the end**. In each one, the move shown is the **only** winning move, and the tempting move, the one a quick player would choose, loses.

How to read the diagrams



Left diagram: the position. Solid arrow: the winning move. Dashed arrow: the tempting move.

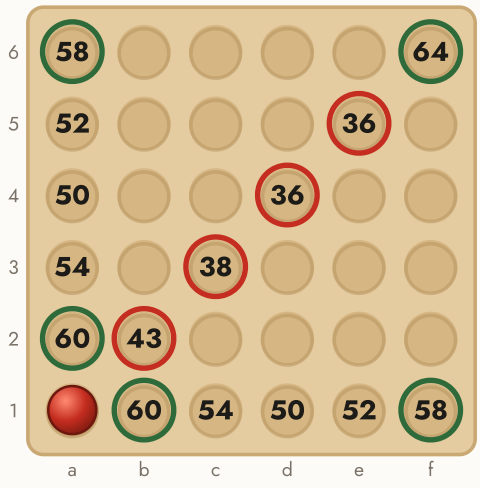


Middle and right: the end of the game after each move, with the best play on both sides. Numbers show the order in which the stones were placed. Green and blue: the largest light and dark groups.

Three questions before every move

1. **Where does my stone fall?** On the square the doron is leaving. You cannot change it, so it is already decided before you move.
2. **What do I give?** The square where I stop the doron receives my opponent's next stone. Does it join one of their groups? Does it link two of them?
3. **What can they give me back?** From that square, my opponent will send the doron along one of its lines, and my next stone will land there. Do those squares touch my groups? Is one of them a dead end?

The seven tactics that follow are answers to these three questions. They come from sorting about 2,500 exactly solved positions in which only one move is best.



The first move. Light's score (%) for each first destination of the doron, from 960 games between strong computer players for each move. Green: best; red: worst.

Strategy 1 · The opening

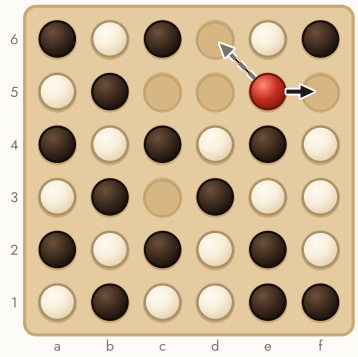
Light's first stone always falls on a1; the real decision is where Dark's first stone will fall.

First move	Light's score
a1 → f6, the opposite corner	64 %
a1 → a2 or b1, along the edge	60 %
a1 → a6 or f1, the other corners	58 %
a1 → b2, c3, d4, e5, the diagonal	36 to 43 %

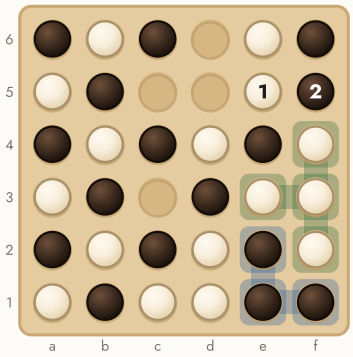
Sending the doron into the centre of the diagonal hands the advantage to Dark: from there, Dark can send it almost anywhere. Corners and edges keep Dark's choices narrow.

TACTIC 1 Lock the doron

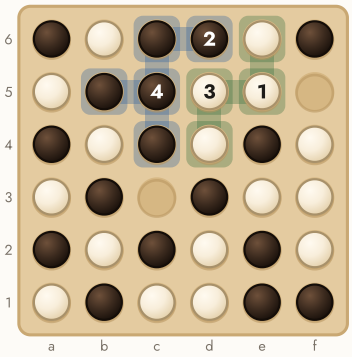
Sending the doron to a square with no way out ends the game: your opponent must take it off and place the last stone in its place. Here Light sends it to **f5**, which is surrounded. Dark's last stone on f5 only joins f6, and the count stops while Light is ahead. Entering the empty pocket instead (d6, d5 or c5) gives Dark the last word there: Dark's stones on d6 and c5 link c6, b5 and c4 into a group of five. *Lock only when the count, as it stands, is yours.*



Light to move. Winning: **f5** (solid). Tempting: **d6** (dashed).



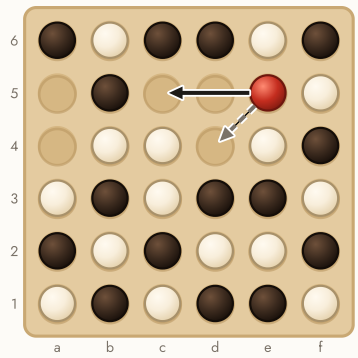
After **f5**: Light **4**, Dark **3**. **Light wins.**



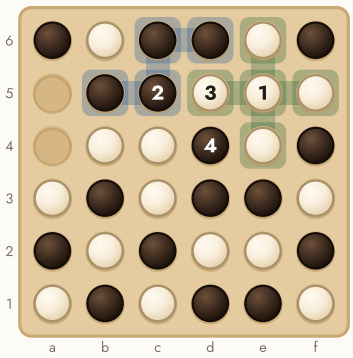
After **d6**: Light **4**, Dark **5**. **Dark wins.**

TACTIC 2 The net

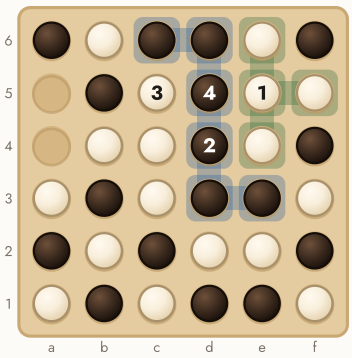
Choose a square from which **every** exit runs into a dead end that suits you. Light is willing to give Dark c5 (Dark's group grows to four) because from c5 the doron has only two exits, d5 and d4, and both are traps: after d5, Light's stone joins its group (5) and the doron is locked on d4; after d4, Light's stone on d4 links eight stones. The natural d4 lets Dark go round through c5 and d5 and finish with six.



Light to move. Winning: **c5** (solid). Tempting: **d4** (dashed).



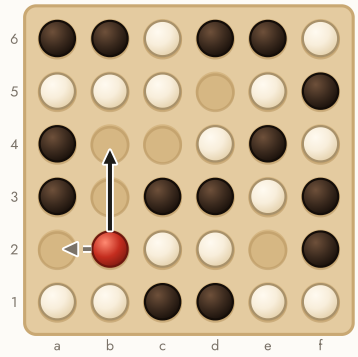
After **c5**: Light **5**, Dark **4**. **Light wins.**



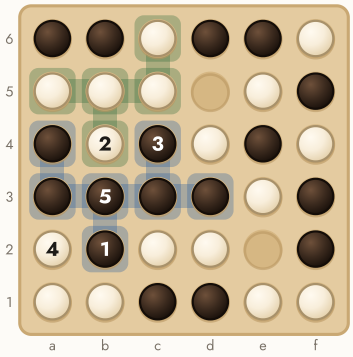
After **d4**: Light **4**, Dark **6**. **Dark wins.**

TACTIC 3 The fork

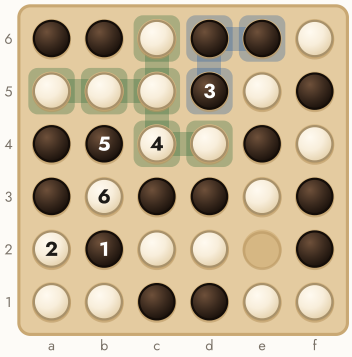
A square where every exit gives you **either** a growing stone **or** a lock. Dark gives Light b4 (Light's group grows to five). From b4, Light can go to b3 or c4. On b3, Dark's stone joins the wall a3–c3–d3 and Dark then locks the doron on a2. On c4, Dark's stone joins the same wall and Dark's last stone on b3 completes a group of seven. The tempting a2 lets Light run the doron across to d5 instead.



Dark to move. Winning: **b4** (solid). Tempting: **a2** (dashed).



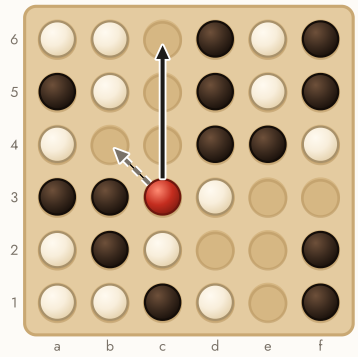
After **b4**: Light **5**, Dark **7**. **Dark wins.**



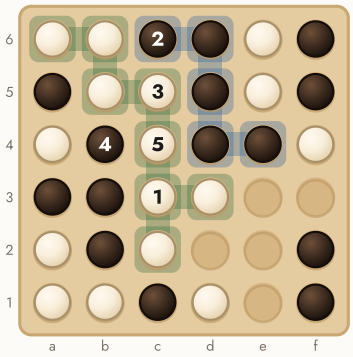
After **a2**: Light **6**, Dark **3**. **Light wins.**

TACTIC 4 Forced junction

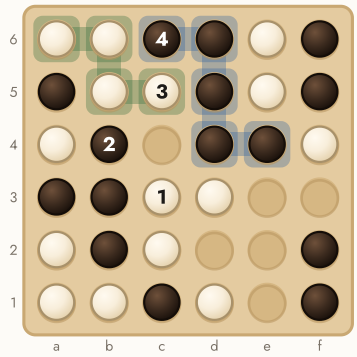
Send the doron where **all** exits touch your own group: whatever your opponent does, your next stone connects. Light gives Dark c6, which joins Dark's group of five. But from c6 the doron can only go down the column, to c5 or c4, and both squares touch light stones. Light's stones then land on c5 and c4 and link the top-left group with c3–d3–c2: **eight** stones. The tempting b4 lets Dark send the doron to c5 at once; the game then ends with the doron locked on c6, c4 stays empty and Light's two groups never meet.



Light to move. Winning: **c6** (solid). Tempting: **b4** (dashed).



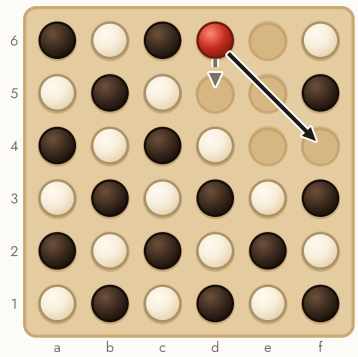
After **c6**: Light **8**, Dark **5**. **Light wins.**



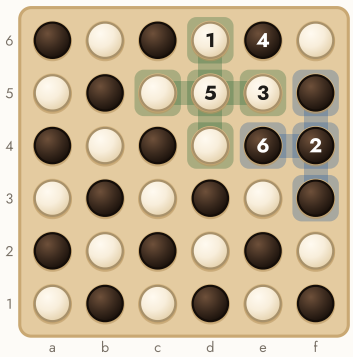
After **b4**: Light **4**, Dark **5**. **Dark wins.**

TACTIC 5 The sacrifice

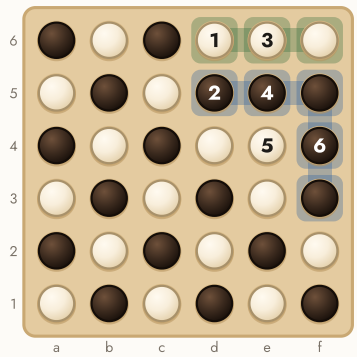
The most frequent winning idea, and the hardest to see: **give your opponent a stone that grows their group**, to control what follows. Every stone here stands alone, except for an empty pocket (d5, e5, e6, e4, f4). Light gives Dark f4, joining f5 and f3 (three stones). In return, the doron must go back into the pocket, and Light's stones fill e5 and d5 to link d6, c5 and d4: Light 5, Dark 4. The quiet-looking d5 gives Dark nothing at first, but Dark's next stones fill d5, e5 and f4 and join f5 and f3: five.



Light to move. Winning: **f4** (solid). Tempting: **d5** (dashed).



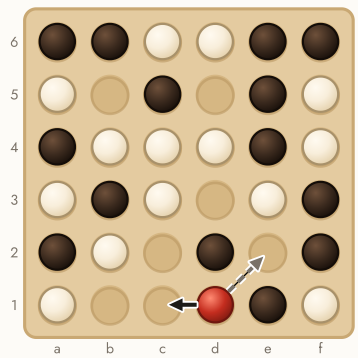
After **f4**: Light **5**, Dark **4**. **Light wins.**



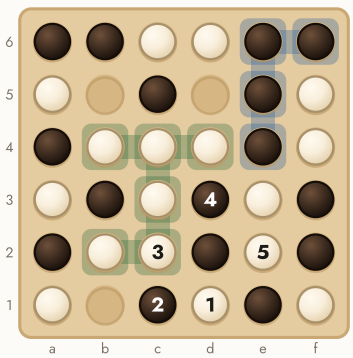
After **d5**: Light **3**, Dark **5**. **Dark wins.**

TACTIC 6 The empty gift

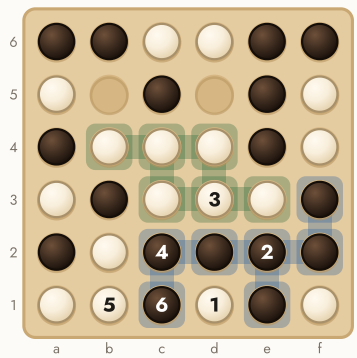
Stop the doron on a square where your opponent's stone will be **useless**, touching none of their stones. Light moves just one square, to c1. Dark's stone will land there, alone; and both of Dark's exits, c2 and b1, touch Light's group, which grows to six. The tempting e2 forces Dark to send the doron to d3, where Light's stone connects at once, but it hands Dark e2, linking four dark stones: Dark finishes with seven.



Light to move. Winning: **c1** (solid). Tempting: **e2** (dashed).



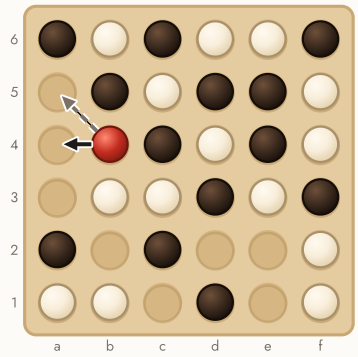
After **c1**: Light **6**, Dark **4**. **Light wins.**



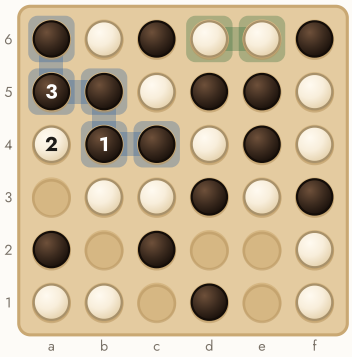
After **e2**: Light **6**, Dark **7**. **Dark wins.**

TACTIC 7 The quiet move

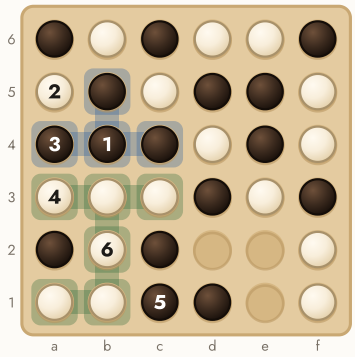
A short move to a square that touches no stone at all can still decide who has the last word. Dark steps into the empty corner, a4. Light must leave by a5 or a3. a5 is a dead end: Dark's last stone lands on a5 and joins a6, b5, b4 and c4 (five). a3 leads to a short race in the bottom-left corner that Dark also wins. The tempting a5 lets Light answer a4; the doron ends up locked on b2, where Light's last stone links six light stones.



Dark to move. Winning: **a4** (solid). Tempting: **a5** (dashed).



After **a4**: Light **2**, Dark **5**. **Dark wins.**



After **a5**: Light **6**, Dark **4**. **Light wins.**

STRATEGY 2 Principles for the whole game

Each principle was measured on about 1,600 positions where the choice matters, then checked on a second set of positions never used to define it. None of them is enough on its own: even the best combination picks an optimal move only 12 points more often than a random choice. Calculation stays indispensable, and that is what keeps every game open.

- **Do not link two of your opponent's groups**, unless you have calculated a sacrifice. In the empty gift above, e2 would have linked four dark stones. *Confirmed in every situation.*
- **Send the doron towards your own groups.** The more of its exits touch your stones, the more likely your next stone connects (the forced junction). *The strongest principle, confirmed in every situation.*
- **When you are ahead, give empty squares** that touch none of your opponent's stones (the empty gift). *Confirmed when ahead; useless when behind.*
- **When you are level or behind, smother:** give the square that blocks your opponent's growth, even if it adds a stone to their group (the sacrifice). *Helps when level or behind, not when ahead.*
- **Lock the doron only when the count, as it stands, is yours** (the closure). Locking just because you are ahead brings nothing.
- **Do not trust simple counting.** Counting the empty squares to guess who will place the last stone gains nothing as a rule of thumb: in a pocket, the last word has to be read out move by move (the net, the quiet move).